Mihir Khatri

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Professional Summary

Undergraduate Computer Science student focused on machine learning systems, GPU-aware inference optimisation, and LLM efficiency research. Published independent researcher with hands-on experience in CUDA kernel development, transformer architectures, and production-grade ML pipelines. Interests span supervised learning, deep neural networks, sequential models, dimensionality reduction, and generative AI with a bias toward measurable, deployment-oriented results.

Education

Sardar Patel Institute of Technology, Mumbai University — Mumbai, India
B.Tech in Computer Science and Engineering — *Expected Aug 2028*

- **GPA:** 9.35 / 10.0
- **Relevant Coursework:** Linear Algebra, Probability & Statistics, Data Structures & Algorithms, Discrete Mathematics, Operating Systems, Computer Networks

Publication

Ro-SVD: Position-Aware SVD Compression of KV Caches in GQA Models — Zenodo, May 2026

- Proposed online activation-level KV cache compression using position-aware truncated SVD on pre-RoPE key tensors in Grouped-Query Attention architectures.
- Evaluated across 1,602 samples on all 16 LongBench subsets using Qwen2.5-7B-Instruct with 4-bit NF4 quantisation on dual NVIDIA T4 GPUs.
- Achieved **73.91% win rate on ROUGE-L** against uncompressed baseline while reducing KV-cache memory by **35.1%**.
- Designed adaptive staircase rank allocation with per-head energy thresholding to preserve quality in high-entropy attention heads.
- Implemented custom CUDA de-rotation kernel with cuBLAS SGEMM batched low-rank projection; derived randomised SVD optimisation path reducing complexity from $\mathcal{O}(Thd^2)$ to $\mathcal{O}(Thr \log r)$.

Industry Experience

Daten & Wissen Pvt. Ltd. — Thane, India
AI/ML Intern — *2025*

- Fine-tuned PaddleOCR models for license plate detection and recognition on a dataset of **2,000 training and 1,000 test images** (cumulative across localiser and recogniser stages), achieving **~88% end-to-end recognition accuracy** after 1.5 days of full training.
- Improved robustness under noisy real-world conditions via geometry correction, adaptive filtering, and contrast normalisation.
- Evaluated segmentation architectures including U-Net, SAM, and ResNet backbones for production-oriented latency–accuracy trade-offs.
- Conducted Optuna hyperparameter optimisation sweeps; analysed convergence stability across training runs.
- Built end-to-end document understanding pipelines using PyTorch and TensorFlow.

Research & Projects

Transformer Encoder-Decoder from Scratch | *PyTorch* — 2025

- Implemented multi-head self-attention, causal masking, positional encoding, and autoregressive decoding from first principles in PyTorch.
- Studied gradient flow, training stability, and architectural behaviour compared with RNN/LSTM baselines; developed intuition for attention complexity and sequence modelling trade-offs.

LLM Inference Optimisation: KV Cache via Low-Rank Approximation | *PyTorch, CUDA, Triton* — 2026

- Built end-to-end pipeline for inverse RoPE de-rotation, truncated SVD, and progressive online re-compression of KV caches.
- Ran 7-configuration evaluation sweep over 1,602 LongBench samples tracking ROUGE-L, perplexity, latency, and GPU memory usage.

- Quantified 29.5% latency overhead of Python-loop SVD; explored Triton-based batched randomised SVD to reduce this cost.
- Demonstrated compatibility between activation-level compression and 4-bit NF4 quantised inference.

OCR + LLM Reconstruction Pipeline | *Multimodal NLU* 2026

- Built multimodal document reconstruction pipeline combining image preprocessing, OCR, and LLM-based reasoning.
- Explored latency–quality trade-offs between local quantised inference and API-based models; analysed hallucination behaviour across prompting strategies.

Raga Classification — Audio Deep Learning | *PyTorch* 2025

- Built a 68-class audio classification system for Indian classical ragas using mel-spectrogram and MFCC representations; achieved **68% classification accuracy**.
- Studied challenges of non-Western musical feature extraction and class imbalance in fine-grained audio taxonomies.

YOLO-Based Rockfall Detection | *Computer Vision* 2025

- Trained YOLO object detection model for safety-critical rockfall detection on limited annotated datasets.
- Analysed precision-recall trade-offs under varying IoU thresholds; explored data augmentation strategies to improve generalisation.

Technical Skills

Languages: Python, C, C++, Java, SQL, CUDA C, JavaScript
ML / DL: PyTorch, TensorFlow, Hugging Face Transformers, PEFT/LoRA, BitsAndBytes, PaddleOCR, YOLO, Ollama
Systems: CUDA kernel development, cuBLAS, Triton, GPU profiling, model quantisation, memory-constrained inference
Tools: Git, GitHub, Kaggle, Jupyter, VS Code, Optuna, LongBench, Google Colab